Criticality of the Hopf bifurcations in the Becker–Döring equations with atomization

chemistry-open-problems · 26 October 2026 · complete for branches 1–3

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Abstract. We answer Pego and Velázquez's stability question for the first three Hopf branches of the Becker–Döring equations with atomization, for every integer maximal cluster size on each branch. The first branch starts at $M = 15$: it is subcritical there, and supercritical with orbitally asymptotically stable periodic solutions in the mass hyperplane for every $M \geq 16$. Branches 2 and 3 start exactly at $M = 44$ and $M = 87$; they are supercritical on the centre manifold, but their periodic solutions are unstable in the full mass hyperplane. The proof combines direct ball-arithmetic certificates for small matrices with a formulation regular at $M^{-1/2} = 0$. A finite cover of this compact parameter interval certifies the first Lyapunov coefficient, simplicity, transversality and the full spectral hypothesis for every $M \geq 161$. Argument-principle counts use interval arithmetic throughout. Certified junctions identify all three curves from their finite onsets to the limiting roots. Exact Hurwitz determinants exclude Hopf bifurcations for $3 \leq M \leq 14$, and symbolic algebra identifies the limit of the rescaled discrete Lyapunov coefficient. This completes the local answer on branches 1–3; no global assertion about distant periodic orbits or all higher branches is made.

1 Problem and source

R. L. Pego, J. J. L. Velázquez, *Temporal oscillations in Becker–Döring equations with atomization*, Nonlinearity 33 (2020) 1812–1846, doi:10.1088/1361-6544/ab6815, arXiv:1905.02605 (v2 read in full, LaTeX source).

Theorem 3.3 of the paper states that for $k \in \mathbb{N}$ and $M > M_{0,k}$ the system admits a Hopf bifurcation as $\kappa = K\sqrt{M}$ passes through each $\kappa_j(M)$, $j = 1, \ldots, k$. Immediately after it the authors write (verbatim, §3.3):

> "We have not managed to determine analytically whether the bifurcating solutions are stable (the supercritical case) or not. Many of our numerical computations, as in Fig. 1, are consistent with the presence of stable periodic solutions, however."

The constant $M_{0,k}$ is not explicit.

Novelty check (2026-10-18): 20 citing works (OpenAlex and Semantic Scholar) were checked; every arXiv full text was grepped. Later work by the authors and searches on key phrases were also checked. Nothing settles the question. The closest work is Niethammer–Pego–Schlichting–Velázquez (SIAM J. Appl. Math. 2022, arXiv:2102.06751). It carries out a *formal* direction-of-bifurcation computation for a *different* model (Becker–Döring with injection and depletion). Details are in `references.md`.

2 Model, linearization, exact quadratic structure

The unknowns are $n_\ell \geq 0$, $1 \leq \ell \leq M$, with $N = M - 1$ and $J_\ell = n_\ell n_1 - n_{\ell+1}$ for $1 \leq \ell \leq N$ (paper (1)–(4)):

$$\dot{n}_\ell = J_{\ell-1} - J_\ell \ (2 \leq \ell \leq N), \qquad \dot{n}_M = J_N - K n_M, \qquad \dot{n}_1 = -J_1 - \sum_{\ell=1}^{N} J_\ell + MK n_M.$$

The mass $\sum_\ell \ell n_\ell$ is conserved.

The constant state $n_\ell \equiv A := 1 + K$ is an equilibrium. Write $n = A\mathbb{1} + v$. A direct expansion (checked in `scripts/01_check_model.py`, claims M1–M3) gives the **exact** identity

$$\dot{v} = Bv + v_1 \, Sv, \tag{2.1}$$

with no cubic or higher terms. Here B is the paper's matrix (27). The linear map S is

$$(Sv)_1 = -2v_1 - \sum_{\ell=2}^{N} v_\ell, \qquad (Sv)_\ell = v_{\ell-1} - v_\ell \ (2 \leq \ell \leq N), \qquad (Sv)_M = v_N.$$

Equivalently, $f(n) = (n_1 - 1)\, Sn + Tn$ with T linear. The nonlinearity is the product of the "transport speed" $c = n_1 - 1$ and a fixed linear operator.

Mass conservation: $\boldsymbol{\ell} = (1, 2, \ldots, M)$ satisfies $\boldsymbol{\ell} B = 0$ and $\boldsymbol{\ell} S = 0$. Hence the hyperplane $H = \{\boldsymbol{\ell} \cdot v = 0\}$ is invariant under (2.1), and $\operatorname{Ran} B = H$.

By paper Lemma 3.1, 0 is a simple eigenvalue of B, with eigenvector $\bar{v} \notin H$. Hence $B|_H : H \to H$ is invertible.

Kuznetsov's multilinear forms for (2.1) are

$$\mathcal{B}(x, y) = x_1 Sy + y_1 Sx, \qquad \mathcal{C} \equiv 0.$$

3 Hopf setting and first Lyapunov coefficient

Setting. As in the paper, the bifurcation parameter is κ, and the dynamics is restricted to the invariant affine hyperplane through $A\mathbb{1}$. In the coordinate v this is H, a fixed space, and the equilibrium is $v = 0$ for every κ.

Eigenvalues. The eigenvalues of $B|_H$ are those of B except the simple eigenvalue 0. At a Hopf point κ_j, B has eigenvalues $\pm i\omega$ with $\omega > 0$, and $\operatorname{Re} d\lambda/d\kappa > 0$ (paper (33), re-certified in §5).

Coefficient. Let $Bq = i\omega q$, let u be a row vector with $uB = i\omega u$, normalized by $uq = 1$, and set $\langle p, x \rangle := ux$. Kuznetsov's formula (*Elements of Applied Bifurcation Theory*, 3rd ed.; the general formula is also given in [1]), with $\mathcal{C} = 0$, reads

$$\ell_1 = \frac{1}{2\omega} \operatorname{Re}\Big[-2\, u\, \mathcal{B}(q, w_{11}) + u\, \mathcal{B}(\bar{q}, w_{20}) \Big],$$

$$w_{11} = (B|_H)^{-1} \mathcal{B}(q, \bar{q}) \in H, \qquad w_{20} = (2i\omega - B)^{-1} \mathcal{B}(q, q). \tag{3.1}$$

Why the full-space objects may be used.

- $q \in H$, because $\boldsymbol{\ell} q = \boldsymbol{\ell} B q / (i\omega) = 0$.

- $\mathcal{B}(x,y) \in H$ for all x, y, because $\boldsymbol{\ell} S = 0$.
- $u|_H$ is the adjoint eigenvector of $B|_H$.
- $w_{20} \in H$ automatically, because H is invariant under B and $2i\omega \notin \operatorname{spec} B$.
- w_{11} must be taken as the unique solution of $Bw = \mathcal{B}(q, \bar{q})$ **in** $\boldsymbol{H}$. Adding a multiple of $\bar{v}$ would change ℓ_1.

Formula (3.1) is basis-free (projection method), so it is the first Lyapunov coefficient of the restricted system.

Sign convention. ℓ_1 depends on the normalization of q only through a positive factor. Two normalizations are used: $q^H q = 1$ ("unit") and $q_1 = 1$.

Since $\operatorname{Re} \lambda'(\kappa_j) > 0$:

- $\ell_1 < 0$ means **supercritical**: periodic orbits exist for $\kappa > \kappa_j$, $\kappa \approx \kappa_j$, with amplitude $\propto (\kappa - \kappa_j)^{1/2}$, and they attract within the centre manifold.
- $\ell_1 > 0$ means **subcritical**.

For $j = 1$, when all other eigenvalues of $B|_H$ have negative real part (hypothesis H2 and the count of §5), a supercritical orbit is orbitally asymptotically stable within H. For $j \geq 2$ the equilibrium already has $j - 1$ unstable pairs, so the bifurcating orbits are unstable in the full system whatever the sign. The criticality is then a centre-manifold statement.

Remark (intrinsic). ℓ_1 depends only on the vector field at the critical point. The conclusion therefore holds for any one-parameter path through the Hopf point in the (K, mass) plane that crosses transversally. An example is K varying at fixed total mass, where the equilibria are the non-constant ones of paper (15).

4 Exact closed-form evaluation of $\boldsymbol{\ell_1}$ for any M

All vectors occurring in (3.1) have the form

$$v_\ell = \sum_e c_e \, \beta_e^\ell + \sum_{k \in P} d_k \, \delta_{\ell k}, \qquad P \subset \{1, 2, N, M\}, \tag{4.1}$$

with finitely many bases β_e. Each base is a monomial $\varphi^{e_0} \bar{\varphi}^{e_1} A^{e_2} s^{e_3} \bar{s}^{e_4}$ with integer exponents; φ and s are defined below. All operations on such vectors are exact and cost $O(1)$ in M. Implementation: `scripts/bdlib.py`, classes `Problem` and `Seq`.

Right eigenvector (paper §5). For $\lambda = (A - \varphi^{-1})(\varphi - 1)$ and $\psi = 1/(A\varphi)$:

$$q_\ell = c_1 \varphi^{M-\ell} + c_2 \psi^{M-\ell}, \qquad (c_1, c_2) = (f(\psi), -f(\varphi)), \qquad f(\varphi) = A\varphi^{M-1} + 1 - \varphi^{-1},$$

valid for $1 \leq \ell \leq M$ whenever φ is a root of the paper's F (eqs. (71), (76)).

Left eigenvector. The transposed equations, columns $2 \leq k \leq N$ of $uB = \lambda u$, read

$$u_{k-1} - (A + 1 + \lambda) u_k + A u_{k+1} = K u_1.$$

Hence $u_k = a_1 \varphi^k + a_2 \psi^k + u_c$, with $u_c = -K u_1 / \lambda$ and $u_1 = (a_1 \varphi + a_2 \psi)\lambda/(\lambda + K)$. Columns 1 and M give a singular 2×2 system for (a_1, a_2); one row of it fixes the null vector.

Linear maps. B and S act on a term β^ℓ as multiplication by $A/\beta - (A + 1) + \beta$ (bulk rows of B) or $1/\beta - 1$ (bulk rows of S). Rows 1 and M, and the neighbours of point

corrections, are evaluated exactly. The full sums in row 1 are geometric sums. The differences are stored as point corrections.

Solving $(B-\mu)w = r$ for r of the form (4.1) (here $\mu = 0$ or $2i\omega$). Ansatz:

$$w = \sum_e \frac{c_e}{A/\beta_e - (A+1) + \beta_e - \mu}\ \beta_e^\ell + h_1 s_1^\ell + h_2 s_2^\ell + e_1\delta_{\ell 1} + e_M\delta_{\ell M},$$

where $s_{1,2}$ are the roots of $s^2 - (A+1+\mu)s + A = 0$:

- for $\mu = 0$: $s \in \{1, A\}$;
- for $\mu = 2i\omega$: one new generator s, the other root being A/s.

The bulk rows $3 \le \ell \le N-1$ hold identically. Rows $2, N$ (which involve w_1, w_M) and rows $1, M$ give four linear equations for (h_1, h_2, e_1, e_M).

Why this family contains the true solution. Rows $2, \dots, N$ form a three-term recurrence. Their solution set is an affine space of dimension 2, and it is the 2-dimensional subfamily of the ansatz fixed by rows 2 and N.

The case $\mu = 0$. Row 1 is redundant, since $\boldsymbol{\ell}\,(Bw - r) = 0$ whenever rows $2, \dots, M$ hold. It is replaced by the condition $\sum_\ell \ell w_\ell = 0$, which selects $w_{11} \in H$. The 4×4 system is solved in ball arithmetic (`acb_mat.solve`). Success proves that it is nonsingular, so the computed ball contains the exact solution. The resulting point corrections are numerically zero ($\sim 10^{-35}$), as they should be when r has no correction at rows 2 and N.

Inner products. $\sum_{\ell=1}^{M} u_\ell v_\ell$ is computed with exact geometric sums $\sum \beta^\ell = (\beta^{M+1} - \beta)/(\beta - 1)$. A base that is identically 1 (zero exponent vector, e.g. $\psi \cdot A\varphi$) contributes the exact count M. Products β^ℓ of size $A^{\pm M}$ are harmless in Arb's arbitrary-exponent balls.

Cross-checks.

- The closed form agrees with a dense floating-point implementation of (3.1) to 10^{-12}–10^{-14} relative error for $M = 20, \dots, 200$, $j \le 4$ (`01_check_model.py`, M6).
- The eigenvector residuals are at rounding level.

5 Certification

Hopf points. Unknowns $\kappa = K\sqrt{M}$ and $z = M(\varphi - 1)$ (the paper's scaling). The system is

$$Q^\varepsilon(z;\kappa) := K^{-3}F(1 + z/M) = 0 \text{ (complex)}, \qquad \operatorname{Re}\Lambda = 0, \quad \Lambda = M\lambda/K.$$

This is 3 real equations in 3 real unknowns, with $F = f_1 g_2 - g_1 f_2$ exactly as in the paper, eqs. (71)–(76).

- Derivatives come from forward-mode dual numbers over Arb balls.
- The Krawczyk operator $\mathcal{K}(X) = m - Yf(m) + (I - YJ(X))(X - m)$ is evaluated on a box X of relative radius 10^{-25}. The inclusion $\mathcal{K}(X) \subset \operatorname{int} X$ proves existence and uniqueness of a zero in X.
- By paper Prop. 5.3 a zero is an eigenvalue $\lambda = (A - \varphi^{-1})(\varphi - 1)$ of B, which is purely imaginary at the zero ($\operatorname{Re}\Lambda = 0$). It is not spurious, since $\operatorname{Im}\varphi \neq 0$.

Nondegeneracy (certified at every point).

- *Simplicity:* $0 \notin \partial_z Q^\varepsilon$ on the box. Then φ is a simple root of F, and by paper Lemma 5.4 $i\omega$ is a simple eigenvalue of B.
- *Transversality:* along the root curve,

$$\mathrm{Re}(\Lambda + \kappa\, d\Lambda/d\kappa) > 0, \qquad dz/d\kappa = -\partial_\kappa Q^\varepsilon / \partial_z Q^\varepsilon.$$

  The first quantity equals $M^{3/2}\,\mathrm{Re}\, d\lambda/d\kappa$.

Sign of ℓ_1. The closed form of §4 is evaluated with $K = \kappa/\sqrt{M}$ and $\varphi = 1 + z/M$ given as the Krawczyk enclosure. The formulas are exact identities at the true Hopf point, so the output ball contains the true ℓ_1. Precision is 192 bits. The Krawczyk boxes have relative radius 10^{-25}, and the resulting ℓ_1 balls relative radius 10^{-36}–10^{-29} (measured at $M = 15$ and $M = 10^5$). The sign is therefore never in doubt.

Remaining hypothesis H2 (no other eigenvalue of $B|_H$ on $i\mathbb{R}$).

- For $M \le 120$ it is certified with Arb's rigorous eigensolver (`acb_mat.eig`, `07_spectrum_H2.py`). All M eigenvalues of $B(K)$, with K the certified ball, are enclosed and proved simple. Exactly one enclosure contains 0, exactly two meet $i\mathbb{R}$ (at $\pm i\omega$), $2(j-1)$ have positive real part and the rest negative real part.
- This also shows rigorously that exactly $j-1$ conjugate pairs are unstable at $\kappa_j(M)$, so the crossing is the j-th one counted at that parameter value. It does not by itself exclude pairs that cross and re-cross the axis at smaller κ; see the remark on the word "first" in §6.
- Result: all 217 points with $M \le 120$ PASS (106 for $j = 1$, 77 for $j = 2$, 34 for $j = 3$); log in `certificates/logs/07_spectrum_H2.log`. The verifier's own code extends this certificate to $M \le 160$ on all three branches (§11).
- For every $M \ge 161$, H2 is now certified by localization and interval winding counts in Section 13. Consecutive parameter boxes are rigorously linked to the limiting Hopf roots; the earlier floating-point label checks are no longer needed.
- *Simplicity and the paper's Lemma 5.4.* The first step of the proof of Lemma 5.4 (the 2×2 matrix $\mathcal{D}(\varphi)$ is not zero) uses a convexity argument that is valid only for real φ. At the certified points φ is not real. The step is then replaced by a direct check: $f(\varphi) \neq 0$ or $f(1/(A\varphi)) \neq 0$. This check is implicit in the certificate, because the eigenvector q, with coefficients $(c_1, c_2) = (f(\psi), -f(\varphi))$, is normalised by a nonzero ball. The verifier also checked it explicitly on a sample (§11). The rest of the proof of Lemma 5.4 then applies verbatim.

Generation and separate re-check.

- The sweep files are generated by `03_certify_range.py` (continuation in M).
- `04_verify_certificates.py` then re-proves every line from the stored midpoints alone, without continuation: Krawczyk, ℓ_1 enclosure, simplicity, transversality. It also checks that every integer M of the range is present.
- Done for every line of all six files ($3 \times (\approx 10^4 + 9 \cdot 10^4)$ Hopf points). Logs are in `certificates/logs/04_verify_*.log`.

6 Results on the finite range

Theorem A (computer-assisted; certificates in `certificates/`). For every pair (M, j) in

$$\mathcal{S} = \{(M,1) : 15 \le M \le 10^5\} \cup \{(M,2) : 44 \le M \le 10^5\} \cup \{(M,3) : 87 \le M \le 10^5\}$$

the scaled Hopf system has a unique zero $(\kappa_j(M), z_j(M))$ in an explicit box. There:

- $\pm i\omega$ is a simple pair of eigenvalues of B;
- $\operatorname{Re} d\lambda/d\kappa > 0$;
- the first Lyapunov coefficient of the system restricted to the mass hyperplane satisfies

$$\boxed{\ell_1 < 0 \ \text{ for all } (M,j) \in \mathcal{S}\backslash\{(15,1)\}, \qquad \ell_1 > 0 \ \text{ for } (M,j) = (15,1).}$$

Consequences (wording corrected at verification):

- Wherever the remaining hypothesis H2 holds, the Hopf bifurcation at $\kappa_j(M)$ is **supercritical**, except for $M = 15$ where it is **subcritical**.
- *Rigorous range.* H2 is certified for every $M \le 160$ on all three branches (solver: $M \le 120$; verifier, separate code: $M \le 160$, and also $M = 250, 300$). Hence, for every integer $16 \le M \le 160$, the Hopf bifurcation at $\kappa_1(M)$ is supercritical and its periodic orbits are orbitally asymptotically stable within the mass hyperplane. At $M = 15$ it is subcritical.
- *Complete range.* Theorem D in Section 13 supplies H2 for every $M \ge 161$ and extends the coefficient sign to arbitrarily large M. Thus the first-branch stability conclusion holds unconditionally for every $M \ge 16$.
- *The word "first".* The stability of the bifurcating orbits only requires the spectrum at $\kappa_1(M)$: all other eigenvalues in $\operatorname{Re}\lambda < 0$, which is H2 together with the count of §5. That $\kappa_1(M)$ is the *first* value of κ at which the constant equilibrium loses stability is a separate statement. The verifier proved it by exact algebra for $15 \le M \le 57$ (§11): for these M, every crossing of the imaginary axis for $K > 0$ is located and the constant equilibrium is linearly stable on the mass hyperplane for all $0 < \kappa < \kappa_1(M)$. For $M > 57$, "first" is used in the sense of the paper's Theorem 3.2, namely continuation from its first limiting Hopf root. Section 13 certifies that identification for the entire branch; a global assertion for all smaller parameters is not needed here.

Remarks.

- The first branch exists from $M = 15$, at $K \approx 212.9$, $\kappa \approx 824.5$. Branch 2 starts at $M = 44$ ($K \approx 232$) and branch 3 at $M = 87$ ($K \approx 265$). All three integer onsets are exact by AM7 in Section 13; the real curves enter from $K = \infty$ at $M_j^* \in (M_j - 1, M_j)$. Exact Hurwitz determinants exclude every crossing for $3 \le M \le 14$, for every $K > 0$. The original verification also located all crossings for $15 \le M \le 57$: one for $15 \le M \le 43$, two for $44 \le M \le 57$.
- $M = 15$ is a genuine exception. $\ell_1 = +1.1952 \times 10^{-9}$ (unit normalization), certified, with tight radius. The dense floating-point formula gives the same value. BifurcationKit, a separate code with automatic differentiation, reports `SubCritical` there and `SuperCritical` at $M = 16, 17, 30$.

- The scale-free value ℓ_1/M (with $q_1 = 1$) is tiny at $M = 15$ ($+5.8 \times 10^{-10}$), against -6.3×10^{-6} at $M = 16$. This suggests that an analytic continuation in M would have a degenerate (Bautin) point between 15 and 16; that continuation is not computed here. For the integer problem the first branch is subcritical exactly at its first value $M = 15$.

Representative certified values. The unit column uses $q^H q = 1$; the last column is ℓ_1/M with the normalization $q_1 = 1$.

M	j	$\kappa_j(M)$	ω	ℓ_1 (unit)	ℓ_1/M ($q_1 = 1$)
15	1	824.53962734	87.797	$+1.195\times10^{-9}$	$+5.8 \times 10^{-10}$
16	1	83.095251377	8.0421	-1.296×10^{-5}	-6.27×10^{-6}
30	1	9.2400641499	0.34187	-6.723×10^{-3}	
100	1	3.9348822214	2.174e-2	-3.001×10^{-2}	
100	2	15.746601845	0.18808	-2.477×10^{-3}	
100	3	121.37739313	2.1986	-1.033×10^{-5}	
10^4	1	2.0375686900	9.36e-6	-1.542×10^{-3}	-0.011015
10^4	2	3.7981676429		-4.525×10^{-3}	-0.007971
10^4	3	5.5155380850		-3.798×10^{-3}	-0.003637
10^5	1	1.9413727095	2.778e-7	-4.028×10^{-4}	-0.009236
10^5	2	3.3593351327	1.162e-6	-1.443×10^{-3}	-0.007509
10^5	3	4.5144663668	2.465e-6	-1.347×10^{-3}	-0.003525

Full data: `certificates/sweep_j{1,2,3}*.csv.gz` (one line per M).

7 The limit $M \to \infty$

7.1 7.1 Formal limit problem

Notation: $\varepsilon = M^{-1/2}$, $K = \varepsilon\kappa$, $x = \ell/M$, $\tau = \varepsilon^3 t$ (paper (35)). Write $n_1 = 1 + c$, $c = \varepsilon\gamma$, $\gamma = \kappa + \Gamma$.

1. **Bulk.** Exactly, $\dot{n}_\ell = c\,(n_{\ell-1} - n_\ell) + (n_{\ell-1} - 2n_\ell + n_{\ell+1})$ for $2 \le \ell \le N$: upwind transport with speed c plus diffusion. With $n_\ell \approx 1 + \varepsilon\kappa + \varepsilon V(x, \tau)$, at leading order

$$V_\tau + \gamma V_x = 0 \quad (0 < x < 1), \qquad V(0, \tau) = \Gamma(\tau).$$

   The inflow value comes from $n_1 = 1 + \varepsilon\gamma$. Diffusion contributes εV_{xx}.
2. **Boundary layer at $\ell = M$**, of width ℓ-scale $1/c = \sqrt{M}/\gamma$. It is quasi-static: its relaxation time is $O(M) \ll M^{3/2}$. The flux $J_\ell = cn_\ell - (n_{\ell+1} - n_\ell)$ is constant across it and equals Kn_M at $\ell = N$. Solving the recurrence gives $n_\ell = a + C(1+c)^\ell$, where a is the outer value and $C(1 + c)^M = a(\gamma - \kappa)/\kappa$. Its excess mass is $\sum \ell(n_\ell - a) \approx M^2 \varepsilon\, a\, \Gamma/(\kappa\gamma)$.
3. **Monomers** relax on the fast time scale $1/(A(M + 2))$. Their equation is replaced by mass conservation, $\sum \ell(n_\ell - A) = 0$. With $a = 1 + O(\varepsilon)$ this gives

$$\int_0^1 xV\, dx + \frac{\Gamma}{\kappa(\kappa + \Gamma)} = 0.$$

Lagrangian clock. Let $dT = \gamma\, d\tau$. Along characteristics V carries its inflow value, so $V(x,\cdot) = G(T-x)$ with $G(T) = \Gamma$. The problem becomes the scalar *renewal equation*

$$\boxed{\frac{G(T)}{\kappa(\kappa + G(T))} + \int_0^1 s\, G(T-s)\, ds = 0} \tag{R}$$

(R) is autonomous, and its only nonlinearity is the boundary-layer term. A monotone time change preserves orbits, their stability and the side of bifurcation, so criticality can be read on (R).

Check against the paper. With $G = e^{\zeta T}$, linearizing (R) gives $\Delta(\zeta) = \kappa^{-2} + E(\zeta) = 0$, where $E(\zeta) = \int_0^1 s e^{-\zeta s} ds = (1 - e^{-\zeta}(1+\zeta))/\zeta^2$. Since $\zeta^2 e^{\zeta} \Delta(\zeta) = Q(\zeta;\kappa)$, this is exactly the paper's limit function Q (eq. (87)). Its purely imaginary roots are $\zeta = \pm i t_j$, where $\tan t_j = t_j$, $\cos t_j < 0$ and $\kappa_j^{02} = \sqrt{1+t_j^2} - 1$.

7.2 7.2 Landau coefficient of (R) (multiple scales)

Multiply (R) by κ^2:

$$G + \kappa^2 LG - G^2/\kappa + G^3/\kappa^2 + O(G^4) = 0, \qquad (LG)(T) = \int_0^1 sG(T-s)ds.$$

Expand with $\kappa = \kappa_c + \eta^2 \tilde{\kappa}$, $G = \eta G_1 + \eta^2 G_2 + \eta^3 G_3$, $G_1 = \mathcal{A}(\eta^2 T) e^{i\omega T} + \text{c.c.}$, $\omega = t_j$, $D(\zeta) := 1 + \kappa_c^2 E(\zeta)$.

- Order η: $D(i\omega) = 0$.
- Order η^2: $G_2 = \alpha \mathcal{A}^2 e^{2i\omega T} + \text{c.c.} + \beta|\mathcal{A}|^2$, with $\alpha = 1/(\kappa D(2i\omega))$ and $\beta = 2/(\kappa D(0))$.
- Order η^3: solvability of the $e^{i\omega T}$ component, using $L[\mathcal{A}(\eta^2 T) e^{i\omega T}] = e^{i\omega T}(E(i\omega)\mathcal{A} + \eta^2 E'(i\omega)\mathcal{A}')$ and $E(i\omega) = -1/\kappa^2$. Explicitly, using $\kappa^2 = \kappa_c^2 + 2\kappa_c \eta^2 \tilde{\kappa} + \ldots$, the $e^{i\omega T}$ component of the order-η^3 equation reads

$$\kappa_c^2 E'(i\omega)\mathcal{A}' + 2\kappa_c \tilde{\kappa} E(i\omega)\mathcal{A} - \frac{2}{\kappa_c}(\alpha+\beta)|\mathcal{A}|^2\mathcal{A} + \frac{3}{\kappa_c^2}|\mathcal{A}|^2\mathcal{A} = 0.$$

  Here $(G_1 G_2)_{e^{i\omega T}} = (\alpha+\beta)|\mathcal{A}|^2\mathcal{A}$ and $(G_1^3)_{e^{i\omega T}} = 3|\mathcal{A}|^2\mathcal{A}$.

The result is the amplitude equation

$$\mathcal{A}' = \sigma\tilde{\kappa}\mathcal{A} + \ell\,|\mathcal{A}|^2\mathcal{A}, \qquad \sigma = \frac{2}{\kappa^3 E'(i\omega)}, \qquad \ell = \frac{b}{\kappa^4 E'(i\omega)}, \qquad b = \frac{2}{D(2i\omega)} + \frac{4}{D(0)} - 3.$$

Exact evaluation at the Hopf points (`05_limit_coefficient.py`, C1–C3, sympy). Set $R = \sqrt{1+t^2} = 1 + \kappa^2$. The Hopf conditions give $e^{it} = -(1+it)/R$, and then:

$$E'(it) = \frac{1}{R} - i\,\frac{R+1}{tR(R-1)}, \qquad E(2it) = \frac{1-it}{2R^2}, \qquad D(0) = \frac{R+1}{2},$$

$$\operatorname{Re}\big(b\,\overline{E'(it)}\big) = -\,\frac{7R^2 - 34R + 23}{R(R+1)(5R-3)}.$$

The last identity is checked exactly in the rational parametrization $t = (s^2-1)/(2s)$, $R = (s^2+1)/(2s)$.

Signs.

- $\operatorname{Re}\sigma > 0$, since $\operatorname{Re}E'(it) = 1/R > 0$. This is the transversal crossing, consistent with paper Lemma 7.3.
- $7R^2 - 34R + 23 > 0$ iff $R > (17 + 8\sqrt{2})/7 \approx 4.0448$. Every Hopf point has $t_j \geq t_1 > 3.92$ (interval check: $\tan t - t$ is increasing on $(\pi, 3\pi/2)$ and negative at 3.92), hence $R_j > 4.0455$.
- Therefore $\operatorname{Re}\ell_j < 0$ **for every** $\boldsymbol{j}$. Equivalently, the sign is that of $-(7\kappa^4 - 20\kappa^2 - 4)$, negative since $\kappa_j^0 \geq \kappa_{\mathrm{cr}} = 1.8982 > 1.7449$.

Proposition B. Every Hopf bifurcation of the limit equation (R), at $\kappa = \kappa_j^0$, $j \geq 1$, has a negative Landau coefficient. Hence it is supercritical in the multiple-scales sense.

- The algebra is exact.
- The multiple-scales coefficient is the standard first Lyapunov coefficient, up to a positive factor; the rigorous Hopf theory for renewal equations is in Diekmann, Getto and Gyllenberg (2008).
- A direct simulation of (R) confirms the amplitude law. At $\kappa = \kappa_1^0 + \tilde{\kappa}$, a stable cycle forms with amplitude ratio (simulation/prediction) 1.038 at $\tilde{\kappa} = 0.004$ and 1.018 at 0.002 (claim C4b).

Prediction for the discrete $\boldsymbol{\ell_1}$ (heuristic matching of normalizations; confirmed numerically in §8). Normalize $q_1 = 1$, so $v_1 = 2\operatorname{Re}z_{\mathrm{Kuz}}$. Use $v_1 = \varepsilon G$, $T \approx \kappa\tau$ and $\tau = \varepsilon^3 t$. Then $c_1 = \varepsilon\kappa\ell$ and $\omega = \varepsilon^3\kappa t_j$, so

$$\frac{\ell_1(q_1 = 1)}{M} \;\longrightarrow\; L_j := \frac{\operatorname{Re}\ell_j}{t_j} = -\left.\frac{R(R-1)(7R^2 - 34R + 23)}{t\,(R+1)(5R-3)\big((R-1)^3 + R + 1\big)}\right|_{t=t_j}.$$

Values: $L_1 = -0.0093114$, $L_2 = -0.0079410$, $L_3 = -0.0037777$, $L_4 = -0.0021661$, $L_5 = -0.0013977$. Also $L_j \sim -7/(5t_j^2) \to 0^-$.

8 Convergence evidence (Observation C)

At sample points $M = 10^4, 3\cdot 10^4, \ldots, 10^8$ and $j = 1, 2, 3$, the Hopf point and ℓ_1 are certified as in §5 (`06_large_M_samples.py`):

- $\ell_1 < 0$ at every sample point;
- the relative deviation of $\ell_1(q_1 = 1)/M$ from L_j at $M = 10^8$ is 1.3×10^{-3} ($j = 1$), 3.6×10^{-3} ($j = 2$) and 6.7×10^{-3} ($j = 3$);
- the deviation decays like $M^{-0.44}$, consistent with an $O(M^{-1/2}) = O(\varepsilon)$ correction.

So the discrete coefficient appears to converge to the limit value predicted by §7. This is numerical support, not a proof, for both the formal limit and the closed form.

9 Validation of the machinery (`02_validate_l1.py`)

- **V1.** The generic Kuznetsov routine reproduces the Guckenheimer–Holmes coefficient on a planar test system to 10^{-12}.
- **V2.** Direct stiff simulation of the full system at $M = 30$, $K = K_c(1 + \delta)$. A small perturbation converges to a small stable cycle. The n_1 half-amplitude divided by the normal-form prediction $2|q_1|\sqrt{-\mu/(\omega\ell_1)}$ is $1.0096, 1.0047, 1.0024, 1.0012$ for $\delta = 4, 2, 1, 0.5\%$: the ratio tends to 1 linearly in δ.
- **V3.** BifurcationKit 0.8 (Julia) classifies the Hopf points as follows: $M = 15$ SubCritical ($\operatorname{Re}b = +1.22 \times 10^{-7}$); $M = 16$, 17 and 30 SuperCritical. Its K_c

agree with ours to 10^{-9}. This code is separate: reduced coordinates, automatic differentiation, its own normal-form code.

- **V4** (`bk_po_branch.jl`, log in `certificates/logs/bk_po_branch.log`). The branch of periodic orbits from the Hopf point is continued by orthogonal collocation (BifurcationKit).
  - $M = 16$: the branch leaves towards $K > K_c$ with amplitude$^2 \propto K - K_c$: supercritical.
  - $M = 15$: the branch leaves towards $K < K_c$ with amplitude$^2 \propto K_c - K$ (ratios 12.4 vs 12.8): subcritical. It then folds back towards $K > K_c$ at n_M-amplitude ≈ 1.2, a saddle-node of cycles, consistent with the tiny ℓ_1.

10 Scope of the complete result

Theorem A certifies the finite-range coefficient. Theorem D in Section 13 completes all local Hopf hypotheses for every integer M on branches 1–3, proves their exact onsets, and identifies the limit of the discrete coefficient with the closed form of Proposition B. The $M \le 160$ certificates and the compact ε-interval certificates overlap at their junction, leaving no gap in cluster size.

The renewal-equation derivation in Section 7 remains formal and is not used to deduce Theorem D. Stability for all $0 < \kappa < \kappa_1(M)$, or for cycles far from the Hopf point, is not asserted. The companion higher-branches folder treats a separate question and retains its own verification status.

11 Verification

A verifier (Claude Opus 5.5, effort xhigh, fresh context, no access to the solver's reasoning) re-derived the key steps with its own code in `verification/`. The full report is `verification.md`. Verdict: **confirmed with corrections**. The corrections concern the scope of the stability statement, the word "first", the floating-point coverage of H2 and a step of the paper's Lemma 5.4. They are applied in §§5, 6 and 10.

- **Separate dense certificate** (`v02_dense_certify.py`). It does not use `bdlib.py`:
  - reduced coordinates $y = (v_2, \dots, v_M)$ on the mass hyperplane, so that no choice of w_{11} is needed;
  - Jacobian and bilinear form obtained from the vector field of the paper itself (exact central differences and polarisation);
  - Krawczyk operator on the full eigen-system $(\hat{A}(K) - i\omega)q = 0$, $q_k = 1$, in $2M$ real unknowns;
  - Kuznetsov's formula in ball arithmetic.

  Every integer M in $[15, 200]$ ($j = 1$), $[44, 200]$ ($j = 2$), $[87, 200]$ ($j = 3$) passed, and $M = 210, 220, \dots, 300$ on each branch: 487 points. At each point the verifier certified simplicity ($uq \neq 0$ with a nonsingular bordered system), transversality, $\omega > 0$ and the sign of ℓ_1. The enclosures agree with the solver's certificate to the double precision in which it is stored. At $M = 15$ the verifier finds $\ell_1 = +1.1951866439 \times 10^{-9}$ (unit normalisation, radius 9×10^{-45}). The cubic term was computed as a third difference of the field and vanishes. With `acb_mat.eig` on the reduced matrix, H2 and the count of unstable pairs are certified for $M \le 160$ on all branches and for $M = 250, 300$.
- **Exact algebra for small M** (`v07_first_crossing.py`, PARI/GP). Let $P(\lambda, K)$

be the characteristic polynomial of $\hat{A}(K)$ in $\mathbb{Z}[K][\lambda]$. The resultant of the even and odd parts of $P(iy, K)$ locates every $K > 0$ with a nonzero imaginary eigenvalue (rigorous real-root isolation). Exact Routh–Hurwitz counts in between then give the number of unstable eigenvalues. Results:

- $3 \le M \le 14$: no crossing, stable for every $K > 0$;
- $15 \le M \le 43$: exactly one crossing, at the certified K_1;
- $44 \le M \le 57$: exactly two crossings, at K_1 and K_2;
- the constant equilibrium is stable on $(0, K_1)$ in all these cases.

For $M \ge 58$ the computation exceeded the memory allotted.

- **Direction of bifurcation without the Lyapunov formula.**
  - Shooting with variational equations for the small periodic orbits at fixed amplitude (`v09_po_shooting.py`): at $M = 15$ the orbits exist for $K < K_c$, with $(K - K_c)/a^2 \to -3.72 \times 10^{-5}$ (subcritical); at $M = 16$ they exist for $K > K_c$, with $(K - K_c)/a^2 = +4.06 \times 10^{-3}$ (supercritical).
  - Direct simulation at $M = 16$ and 50 (`v08_simulate.py`), for $K = K_c(1 + \delta)$, $\delta = 1$–4%. Perturbations starting inside and outside converge to one cycle. Its amplitude matches $2|q_1|\sqrt{-\mu/(\omega\ell_1)}$ to 2×10^{-4} at $M = 16$, and to 0.4–1.5% ($\propto \delta$) at $M = 50$. For $K = 0.98K_c$ the perturbations decay.

  These tests use none of the Lyapunov-coefficient machinery, so they check its sign convention end to end.
- **Large M.** A sparse $O(M)$ re-implementation (tridiagonal plus rank one in the reduced coordinates):
  - in double precision (`v03_sparse_float.py`), at 84 points (25 random $M \le 10^5$ per branch plus boundary cases): the sign agrees everywhere. The relative agreement is $\le 2 \times 10^{-7}$ for $M \le 10^4$ and $\le 9 \times 10^{-5}$ up to 10^5, the limit of double precision;
  - at 320 bits (`v04_sparse_arb.py`), at 12 points: the onsets $(15, 1)$, $(44, 2)$, $(87, 3)$ and $(16, 1)$; $M = 10^3, 10^4$; three random points; and $M = 10^5$ on each branch. κ agrees with the certificate to 10^{-40}, and ℓ_1 in both normalisations to $\le 10^{-16}$, the float64 storage limit of the certificate files.

  The Hopf point is re-located from a perturbed start in both cases. The solver's own code re-proved 1360 certificate lines (`v05_recheck_sample.py`): all $M \le 300$, 200 random $M \le 10^5$ per branch, and the boundary cases. The extra checks $\omega > 0$, $\operatorname{Im} z > 0$ and $f(\varphi) \ne 0$ or $f(\psi) \ne 0$ all pass. An audit of the six certificate files (`v01_audit_csv.py`) checks coverage, signs, $\omega > 0$ and smoothness and monotonicity in M.
- **Limit problem** (`v06_limit_algebra.py`): the identities of §7.2, the sign for every j and the values L_j are re-derived. The reduction to (R) was re-derived by hand; it remains formal.

12 Reproduction

From the project root in Git Bash:

```
outils/wsl "cd chemistry-open-problems/problems/
becker-doring-atomization-hopf/scripts && python <script>"
```

script	content	runtime
`01_check_model.py`	model identities, paper's eigenvalues and Table 1, closed form = dense	1 min
`02_validate_l1.py`	planar test, stiff simulation (M = 30), BifurcationKit cross-check	10 min
`03_certify_range.py J MMIN MMAX OUT.csv [SEED.csv]`	certified sweep (Theorem A)	12–30 ms per M
`04_verify_certificates.py STRIDE files...`	separate re-check of the sweep files	20 ms per line
`05_limit_coefficient.py`	limit problem, exact algebra, simulation of (R)	1 min
`06_large_M_samples.py`	certified samples up to $M = 10^8$, convergence to L_j	2 min
`07_spectrum_H2.py MMAX files...`	rigorous spectra, H2 and branch label for $M \le$ MMAX	15 min
`bk_check.jl M Kc ...`	BifurcationKit normal form (separate)	2 min
`bk_po_branch.jl`	BifurcationKit periodic-orbit branch at $M = 15, 16$ (direction of bifurcation)	3 min

The verifier's scripts `v01`–`v09` are in `verification/` and are described in §11 and in `verification.md`, with their logs and run times.

13 Completion for every cluster size on branches 1–3

The certificates below remove both restrictions of the original finite-range result: the missing spectral hypothesis for intermediate M and the unbounded range $M > 10^5$. All stability statements concern the invariant mass hyperplane through the constant equilibrium. The term "branch 1" identifies the continuation from the first limiting Hopf root; it does not assert stability for every parameter below that point.

Theorem D (complete local criticality on branches 1–3). For each $j = 1, 2, 3$, the Hopf curve continuing the jth root of the Pego–Velázquez limit characteristic function is a graph over $M \in (M_j^*, \infty)$, with

$$14 < M_1^* < 15, \qquad 43 < M_2^* < 44, \qquad 86 < M_3^* < 87.$$

It has exactly one point at every integer $M \ge M_j$, where $(M_1, M_2, M_3) = (15, 44, 87)$, and none below this onset. At every such point the imaginary pair is simple, crosses transversally towards the right as κ increases, and is the only spectrum on the imaginary axis after restriction to H. The first Lyapunov coefficient is negative except at $(j, M) = (1, 15)$, where it is positive. Consequently:

- Branch 1 has orbitally asymptotically stable small periodic orbits for $\kappa > \kappa_1(M)$ close to the bifurcation, for every $M \ge 16$. At $M = 15$ its small periodic orbits are subcritical and unstable.

- Branches 2 and 3 are supercritical on the two-dimensional centre manifold, for every $M \geq 44$ and $M \geq 87$, respectively. Their small periodic orbits are unstable in H because a transverse unstable eigenvalue is present.
- For $3 \leq M \leq 14$ the equilibrium is linearly asymptotically stable on H for every $K > 0$; there is no Hopf bifurcation.

The uniqueness here is for the specified connected Hopf curves, not a classification of all possible disconnected components of the Hopf set.

13.1 AM1. A characteristic function regular at infinity

Put $\varepsilon = M^{-1/2}$, $K = \kappa\varepsilon$, $A = 1 + K$, $\varphi = 1 + \varepsilon^2 z$. Pego–Velázquez's characteristic determinant has the exact form

$$F = -P_1 + \varphi^M P_2 + A^{-M} R_1 + (A\varphi)^{-M} R_2,$$

where, writing $S_0 = (\varphi - 1)(A\varphi - 1)(A\varphi^2 - 1)$,

$$\begin{aligned} P_1 &= K[(A\varphi - 1)^2\varphi^2 - (\varphi - 1)^2] + (KM + 1)S_0, \\ P_2 &= \varphi(A\varphi - 1)^3 - A^2(\varphi - 1)^2 + MA^2(A\varphi - 1)(\varphi - 1)^2, \\ R_1 &= -A^2[(A\varphi - 1)^2\varphi^2 - (\varphi - 1)^2], \\ R_2 &= A(\varphi - 1)^3 + A^2\varphi(A\varphi - 1)^2[M(\varphi - 1) + \varphi]. \end{aligned}$$

Except at the real exceptional points $1, 1/A, \pm A^{-1/2}$, zeros correspond to eigenvalues $\lambda = (A - \varphi^{-1})(\varphi - 1)$. A simple zero gives a simple eigenvalue when $\varphi f(\varphi) = A\varphi^M + \varphi - 1 \neq 0$, as checked below; this supplies the extra condition in the corrected application of PV Lemma 5.4. The eigenvalue zero of B is simple and is removed on H (PV Lemma 3.1).

Define the analytic functions

$$\mathsf{L}(u) = \frac{\log(1 + u)}{u}, \quad \mathrm{Ex}(b) = \frac{e^b - 1}{b}, \quad \mathrm{Ex}_1 = \mathrm{Ex}', \quad \mathsf{N}_1(u) = \frac{1 - \mathsf{L}(u)}{u}$$

by continuity at zero, and set

$$g = \kappa + A\varepsilon z, \quad h = g\varphi + \varepsilon z, \quad E = \exp\Big(z\mathsf{L}(\varepsilon^2 z)\Big), \quad T = \exp[-(\kappa/\varepsilon)\mathsf{L}(\kappa\varepsilon)],$$
$$\widehat{\Lambda} = gz/\varphi, \qquad \lambda = \varepsilon^3\widehat{\Lambda}, \qquad \Lambda = M\lambda/K = z + \frac{\varepsilon z^2}{\kappa\varphi}.$$

Then $Q^\varepsilon = K^{-3}F = E\widetilde{P}_2 - \widetilde{P}_1 + (T/\varepsilon)W/\kappa^3$, where

$$\begin{aligned} \widetilde{P}_2 &= (\varphi g^3 - \varepsilon A^2 z^2 + A^2 z^2 g)/\kappa^3, \\ \widetilde{P}_1 &= [\kappa(g^2\varphi^2 - \varepsilon^2 z^2) + (\kappa + \varepsilon)zgh]/\kappa^3, \\ W &= -A^2(g^2\varphi^2 - \varepsilon^2 z^2) + [A\varepsilon^4 z^3 + A^2\varphi g^2(z + \varphi)]/E. \end{aligned}$$

Thus $Q^0 = e^z(1 + z^2/\kappa^2) - (1 + z)$. These identities are checked symbolically for indeterminate M, rather than by fitting finitely many matrices (checks F1–F4).

Exponential remainder bound. On $\varepsilon \in [e_-, e_+]$, $\kappa \in [k_-, k_+]$, the quantity T/ε^r is bounded by

$$\frac{\exp(-k_-/e_+ + k_+^2/2)}{e_+^r} \quad (e_+ < k_-/r), \qquad \frac{T(e_+, k_-)}{e_-^r} \quad (e_- > 0).$$

Use the smaller valid bound, with the first condition interpreted without division by zero when $r = 0$. The first follows from $\log(1+x) \geq x - x^2/2$ and monotonicity of $\exp(-c/e)/e^r$. For the second, T increases with e and decreases with k, since $x/(1+x) - 2\log(1+x) < 0$. Moreover $\partial_\kappa(T/\varepsilon) = -T/(\varepsilon^2 A)$ and $|\partial_\varepsilon(T/\varepsilon)| \leq (3\kappa+\varepsilon)T/\varepsilon^3$. All remainder bounds extend by zero at $\varepsilon = 0$. The functions above and their derivatives are enclosed by convergent power series with explicit geometric or factorial tails.

13.2 AM2. Uniform Hopf boxes and their identification

For $X = (\kappa, x, y)$ define $G(X;\varepsilon) = (\operatorname{Re} Q^\varepsilon, \operatorname{Im} Q^\varepsilon, \operatorname{Re}\Lambda)$, $z = x + iy$. Let m and e_m be the midpoints of X and a parameter interval E_0, and let Y be an approximate inverse of the midpoint Jacobian. The parametric Krawczyk enclosure is

$$\mathcal{K} = m - YG(m;e_m) - (Y\partial_\varepsilon G(m;E_0))(E_0 - e_m) + (I - YD_X G(X;E_0))(X - m).$$

If $\mathcal{K} \subset \operatorname{int} X$, every parameter in E_0 has a unique zero in X. Indeed, apply the fundamental theorem of calculus to $YG(m;e)$:

$$YG(m;e) - YG(m;e_m) = \int_{e_m}^{e} Y\partial_\varepsilon G(m;t)\,dt.$$

This justifies forming $Y\partial_\varepsilon G$ before interval multiplication by $e - e_m$. The ordinary Krawczyk theorem then applies for each fixed e. Strict inclusion also bounds the weighted row sums of $I - YD_X G$ below one, ensuring uniqueness and nonsingularity.

On every stored box we also certify

$$0 \notin Q_z, \quad |AE + \varepsilon^2 z| > 0, \quad y > 0, \quad \operatorname{Im}\Lambda > 0, \quad \operatorname{Re}(\Lambda_\kappa - \Lambda_z Q_\kappa/Q_z) > 0.$$

These give simplicity, positive frequency, and transversality. All derivatives are obtained by forward differentiation of the explicit formulas, including the exponential remainders.

The first interval begins at zero. There, the root lies in $(2j-1)\pi < y < (2j-\frac{1}{2})\pi$ and has $\kappa > 0$, identifying the jth root $\tan y = y$, $\cos y < 0$. At every shared parameter endpoint a separate narrow Krawczyk enclosure is contained in both neighbouring boxes. Therefore their zeros coincide, and the entire chain is the continuation of this limiting root. This last step is checked by `check_branch_links.py`; overlap of two large boxes alone would not suffice.

13.3 AM3. Explicit localization of the spectrum

For $w = \varphi - 1$, $a = |w|$, $u = \operatorname{Re} w$, one has

$$\operatorname{Re}\lambda = Au - \frac{u + a^2}{1 + 2u + a^2}\,.$$

Choose the root φ of modulus at least $A^{-1/2}$. If $\operatorname{Re}\lambda \geq 0$, then $|\varphi| \geq 1$ and $\operatorname{Re}\varphi \geq 1$: writing $\varphi = re^{i\theta}$, $\operatorname{Re}\lambda = \cos\theta(Ar + r^{-1}) - (A+1)$; the range $A^{-1/2} \leq r < 1$ makes this negative, and for $r \geq 1$ it implies $r\cos\theta \geq (A+1)r^2/(Ar^2+1) \geq 1$. Consequently $u \geq u_{\min}(a)$, the positive root of $2Au^2 + (K + Aa^2)u - a^2 = 0$, increasing in a and decreasing in K.

Here are sufficient inequalities, with explicit constants, to exclude every root with $a \geq R_0/M$. For $a \leq a_1 = 0.1$, put $\mathcal{T} = MA^2a^2|K + Aw|$. The triangle inequality in F,

divided by $\mathcal{T}$, excludes a zero if $\rho_a + \cdots + \rho_f < 1$, where

$$\begin{aligned}
\rho_a &= (1+a_1)(\kappa/(AR_0)+\varepsilon)^2, & \rho_b &= \varepsilon/\kappa,\\
\rho_c &= (1+a_1)^2[\kappa^2/(A^2R_0^2)+\kappa\varepsilon/(AR_0)], & \rho_d &= \varepsilon^2/A^2,\\
\rho_e &= \frac{\kappa^2+\kappa\varepsilon}{A^2R_0}+\frac{(\kappa\varepsilon+\varepsilon^2)(2+a_1)}{A}, & \rho_f &= \frac{N_0T\varepsilon^{-3}+N_1T\varepsilon^{-5}}{A^2R_0^2\kappa},
\end{aligned}$$

with

$$\begin{aligned}
N_0 &= A^2[(K+Aa_1)^2(1+a_1)^2+a_1^2]+Aa_1^3+A^2(1+a_1)^2(K+Aa_1)^2,\\
N_1 &= A^2(1+a_1)(K+Aa_1)^2a_1.
\end{aligned}$$

For example, $|\varphi||K+Aw|^3/\mathcal{T} \le \rho_a$ follows from $a \ge R_0/M$; the $(KM+1)S_0$ term gives ρ_e. The remaining terms follow by $|K+Aw| \ge \max(K, Aa)$, $|\varphi| \le 1+a$, and $|(A\varphi)^{-M}| \le T$.

For larger a, subdivide into $[a_-, a_+]$ and verify

$$(1+2u_{\min}(a_-)+a_-^2)^{M_{\mathrm{lo}}/2}\underline{|P_2|/M} > \overline{|P_1|/M}+\overline{T(|R_1|+|R_2|)/M}, \qquad M_{\mathrm{lo}} = e_+^{-2}.$$

The explicit bounds used are

$$\begin{aligned}
|P_2| &\ge MA^2a^2|K+Aw|-|\varphi||K+Aw|^3-A^2a^2,\\
|P_1| &\le K(|K+Aw|^2|\varphi|^2+a^2)+(KM+1)a|K+Aw|(K+2Aa+Aa^2),\\
T(|R_1|+|R_2|) &\le T\{A^2(|K+Aw|^2|\varphi|^2+a^2)+Aa^3+A^2|\varphi||K+Aw|^2(Ma+|\varphi|)\}.
\end{aligned}$$

When $e_- > 0$, these bounds can also cover $[R_0e_-^2, a_1]$ instead of the six-ratio test.

For branch 1, $K < 1$ on all boxes. The first Gershgorin disc lies strictly in the left half-plane because its centre-to-radius margin is $M(1-K)+6K > 0$. The other discs give $|\lambda| \le 2A+2$. Since $|\lambda| \ge Aa^2/(1+a)$, it suffices to check $a \le a_{\max} = 2+2\sqrt{2}$. For branches 2 and 3 no restriction on K is needed. With $r = |\varphi|$ one may replace the lower bound for $|P_2|/M$ by the maximum of that bound and

$$A^3r^3-\frac{3A^2r^3+(3A+A^2)r^2}{M}-A^2(1+2A)r^2-A^2(2+A)r-\frac{(1+2A^2)r}{M}-A^2(1+M^{-1}).$$

This follows by expanding P_2 and using $|\varphi+M| \ge M+1$. The finite subdivision ends at $a_{\mathrm{big}} = 12$. For $r \ge r_b = a_{\mathrm{big}}-1$, set $\alpha = K/2+3A/2$, $\beta = K/4+15A/4$, and $d = M^{-1}$. Bounds for the tail are

$$\begin{aligned}
C_1 &= K(\alpha^2+9/16)d+\tfrac{3}{2}(K+d)\alpha\beta,\\
C_2 &= T\{[A^2(\alpha^2+9/16)+27A/16+A^2\alpha^2]d+\tfrac{3}{2}A^2\alpha^2\},\\
c &= A^3-3A^2d-(3A+A^2)d/r_b-A^2(1+2A)/r_b\\
&\quad -A^2(2+A)/r_b^2-(1+2A^2)d/r_b^2-A^2(1+d)/r_b^3.
\end{aligned}$$

Take lower endpoints for positive terms in c and upper endpoints for subtracted terms. Then $|P_1|/M \le C_1r^4$, $T(|R_1|+|R_2|)/M \le C_2r^4$, and $|P_2|/M \ge cr^3$. The certified inequality $r_b^{M_{\mathrm{lo}}-1}c > C_1+C_2$ excludes the entire tail. Thus all possible unstable or imaginary eigenvalues are localized to $|z| < R_0$.

13.4 AM4. Counting zeros without floating-point angles

The following argument-principle calculation uses balls throughout. On a contour edge let B_e enclose Q^ε for the whole edge and parameter box. At the midpoint parameters let v_0, v_1 enclose the endpoint values. Subdivide until $\mathrm{Re}(B_e/v_0) > 0$. The actual edge image, rotated by its initial value, lies in the right half-plane; its argument increment is therefore $\arg(v_1/v_0)$, computed in Arb. The sum divided by 2π must contain a unique integer. Nonvanishing on all edges and homotopy invariance make this count valid throughout the parameter box. The former floating-point sum of angles is not used.

For branch 1 take $R_0 = 8$, $\sigma = 0.3$, and a convex polygon enclosing $D = \{|z| \le R_0,\ \mathrm{Re}\, z \ge -\sigma\}$. The polygon's supporting half-planes are checked in ball arithmetic, including the slight outward displacement of its left chord. If $x \le -\sigma$ and $|z| \le R_0$,

$$\mathrm{Re}\,\Lambda \le \max(-\sigma + \varepsilon\sigma^2/\kappa, -R_0 + \varepsilon R_0^2/\kappa) + \frac{\varepsilon^3 R_0^3}{\kappa(1-\varepsilon^2 R_0)} < 0.$$

The bound follows by expanding $1/\varphi$ about one and maximizing the resulting convex quadratic in x. Each box has winding count four. Two zeros are $z = 0$, counted at least twice because $F(1) = F'(1) = 0$; the other two are the certified Hopf pair. Localization and the displayed inequality leave no other eigenvalue with nonnegative real part on H.

For branches 2 and 3 use polygons following level curves of $\mathrm{Re}\,\Lambda$. The constants are respectively

$$(R_0, Y, c_L, c_R) = (20, 20.4, -0.3, 0.25), \qquad (26, 26.6, -0.15, 0.25).$$

Piecewise linear left and right graphs $\ell(y), r(y)$ have strictly negative and positive $\mathrm{Re}\,\Lambda$, respectively, certified edge by edge. They lie in $\mathcal{R} = [x_{\min}, R_0] \times [-Y, Y]$. On this rectangle, with $\rho = \max_{\mathcal{R}} |z|$, the check

$$\frac{\varepsilon}{\kappa}\,\frac{\rho(2+\varepsilon^2\rho)}{(1-\varepsilon^2\rho)^2} < 1$$

implies $\partial_x \mathrm{Re}\,\Lambda > 0$. To exclude the part of $|z| \le R_0$ left of $x_{\min} = -\sigma$, put $q = \varepsilon^2 R_0 \le 2/5$ and $g_0(x) = x + \varepsilon x^2/\kappa + \varepsilon^3|x|^3/[\kappa(1-q)]$. The sharper bound is $\mathrm{Re}\,\Lambda \le \max(g_0(-\sigma), g_0(-R_0)) < 0$. Indeed the upper bound at fixed x, as a function of $t = y^2$, is $x + (\varepsilon/\kappa)(x^2 - t) + \varepsilon^3(x^2+t)^{3/2}/[\kappa(1-q)]$; its derivative is nonpositive when $q \le 2/5$, and g_0 is convex for $x < 0$.

The polygons $P = \{\ell(y) < x < R_0, |y| < Y\}$ and $P_- = \{\ell(y) < x < r(y), |y| < Y\}$ have counts $2j+2$ and four. Localization and monotonicity imply that all nonnegative spectrum is in P, and all imaginary spectrum is in P_-. Its four zeros are exactly $0, 0, z_H, \bar{z}_H$. The remaining $2j-2$ zeros in $P \backslash P_-$ give eigenvalues with positive real part; the real exceptional points of F lie outside this region. This proves H2 and instability. No claim about simplicity of these additional zeros is needed.

13.5 AM5. A regular formula for the nonlinear coefficient

Let $q_1 = 1$, $pB = i\omega p$, $pq = 1$, $r(\mu) = (\mu - B)^{-1} Sq$, where $r(0) \in H$, and $\sigma(v) = pSv$. The quadratic projection formula of Section 3 gives exactly

$$\ell_1 = \omega^{-1}\,\mathrm{Re}\,\big[\sigma(r(0)) + \sigma(\overline{r(0)}) + 2\,\mathrm{Re}(r(0)_1)\sigma(q) + \sigma(r(2i\omega)) + r(2i\omega)_1\sigma(\bar{q})\big].$$

For example $w_{11} = -(r(0) + \overline{r(0)})$, $w_{20} = 2r(2i\omega)$, which proves this identity by substitution. In computing a resolvent, the bulk rows, last row and $\sum_l l r_l = 0$ replace the first row: multiplying by the conserved mass row proves the latter automatically for $\mu \neq 0$, and selects the restricted inverse for $\mu = 0$.

Here are all the blocks needed to evaluate the formula without singular cancellation. Write $T_q = T/\varepsilon$ and

$$C_1 = \frac{g - A^2\varphi T_q/E}{gE/\varphi + \varepsilon^2 z A T_q/E}, \qquad \widehat{C}_2 = \frac{AE + \varepsilon^2 z}{gE + \varepsilon^2 z A\varphi T_q/E},$$
$$q_l = C_1 E\varphi^{-l} + \varepsilon^{-1}\widehat{C}_2 (A\varphi)^{l-M}.$$

For the unnormalized adjoint put $\psi = (A\varphi)^{-1}$, $\theta = \widehat{\Lambda}/(\kappa + \varepsilon^2\widehat{\Lambda})$, $c_A = (\kappa+\varepsilon)\theta + \kappa$, and

$$a_1 = c_A\psi + \varepsilon^2 T_q(Az - \varepsilon\widehat{\Lambda})/E,$$
$$a_2 = -c_A\varphi + E(g/\varphi + \varepsilon^2\widehat{\Lambda}),$$
$$p_l = a_1\varphi^l + a_2\psi^l - (1 - \varepsilon^2\theta)(a_1\varphi + a_2\psi).$$

Divide by $N_p = \sum_l p_l q_l$ when forming σ. For $l < M$,

$$(pS)_l = a_1\varepsilon^2 z\varphi^l - a_2\frac{\varepsilon g}{A\varphi}\psi^l - \varepsilon^2\theta(a_1\varphi + a_2\psi), \qquad (pS)_M = 0.$$

These formulas follow by substitution in the second-order bulk recurrence and its boundary rows; the nonsingular normalization is checked by the ball evaluation.

For $\mu = \varepsilon^3 m$, where $m = 0$ or $2\widehat{\Lambda}$, put

$$c_m = \kappa + \varepsilon^2 m, \quad \zeta = \frac{-2m}{c_m + \sqrt{c_m^2 + 4\varepsilon m}}, \quad s_1 = 1 + \varepsilon^2\zeta, \quad s_2 = A/s_1,$$

and $\Xi = (\kappa - \varepsilon\zeta)/(1 + \varepsilon^2\zeta) + A\varepsilon z - \varepsilon^2 m$. Then

$$r_l = \frac{zC_1E}{\varepsilon(m - \widehat{\Lambda})}\varphi^{-l} + \frac{\widehat{C}_2 g}{\varepsilon\Xi}\frac{(A\varphi)^{l-M} - s_2^{l-M}}{A\varphi - s_2} + \frac{\widehat{h}_1}{\varepsilon}s_1^l + \frac{\widehat{H}_2}{\varepsilon^2}s_2^{l-M}.$$

The two coefficients are obtained from the last row and the mass condition, scaled by ε and ε^5 respectively. These are explicit two-by-two linear equations. The divided difference follows from $(w - s_2)(w + s_2 - A - 1 - \mu) = -(\mu - \lambda)w$, $w = A\varphi$; it removes the subtraction of two individually larger boundary layers.

For completeness the finite sums used in these blocks are evaluated as follows. For $\beta = 1 + \varepsilon^2 b_0$, $b = M\log\beta = b_0\mathsf{L}(\varepsilon^2 b_0)$, and $\gamma = A(1 + \varepsilon^2 g_0)$, geometric summation gives

$$\sum_{l=1}^{M-1}(\beta_1\beta_2)^l = \varepsilon^{-2}[\mathrm{Ex}(b_1 + b_2)\mathsf{L}(\beta_1\beta_2 - 1) - \varepsilon^2],$$
$$\sum_{l=1}^{M-1}\beta^l\gamma^{l-M} = \frac{e^b - \beta\gamma^{1-M}}{\beta\gamma - 1},$$
$$\sum_{l=1}^{M} l\beta^l = \varepsilon^{-4}\beta[\mathsf{L}(u)^2\mathrm{Ex}_1(b) + \varepsilon^2 e^b\mathsf{N}_1(u)], \quad u = \beta - 1,$$
$$\sum_{l=1}^{M} l\gamma^{l-M} = \varepsilon^{-3}(\gamma/d_\gamma - \varepsilon\gamma/d_\gamma^2) + \gamma^{1-M}/(\gamma-1)^2, \quad d_\gamma = (\gamma - 1)/\varepsilon.$$

Divided differences of these identities give the corresponding sums with $(\gamma_1^{l-M} - \gamma_2^{l-M})/(\gamma_1 - \gamma_2)$; their regular parts are $-\beta e^b/(\varepsilon^2 d_1 d_2)$ for the pair sum, where $d_i = (\beta\gamma_i - 1)/\varepsilon$, and

$$\varepsilon^{-4}\left[-\frac{1-\varepsilon^2}{d_1 d_2} + \frac{\varepsilon(d_1+d_2)}{d_1^2 d_2^2}\right]$$

for the mass sum, now with $d_i = (\gamma_i - 1)/\varepsilon$. The remainders are bounded by the supremum of the derivative of the exponentially small part along the segment $[\gamma_1, \gamma_2]$. Products with the layer ψ'^l, $\psi' = (1+\varepsilon^2 p_0)/A$, use $\sum_{l=1}^{M-1}(\psi'\beta)^l = \psi'\beta[1-(\psi'\beta)^{M-1}]/(1-\psi'\beta)$; products with a layer at M are exponentially small. The bound in AM1 controls all remainders, including coincident bases. Direct summation and dense linear solves check these identities entrywise (F5–F6), without using the earlier closed-form implementation.

Track a coefficient as $v\varepsilon^p$, with integer p and regular v. The five terms in the bracket above have order ε; all negative powers cancel before interval evaluation. Thus

$$\widehat{L} = \varepsilon^2 \ell_1(q_1 = 1) = \frac{\mathrm{Re}(\varepsilon^{-1}\mathrm{bracket})}{\mathrm{Im}\,\widehat{\Lambda}}$$

is continuous at $\varepsilon = 0$. To control interval overestimation, regard each exponential remainder as an independent variable ν_i in a disk valid on the whole box. For the resulting regular expression Φ, at any point ξ of the box with midpoint c,

$$\Phi(\xi; \nu(\xi)) \in \Phi(c; \mathcal{D}) + D_\xi \Phi(\mathrm{box}; \mathcal{D})(\xi - c).$$

This is the mean-value formula at fixed $\nu(\xi)$, not differentiation of a numerical remainder bound. The same disks are recorded and replayed in the box and midpoint evaluations.

13.6 AM6. Finite certificates covering the infinite range

All computations use python-flint 0.9 at 192 bits, except the higher precision limiting evaluation. The exact rational endpoint stored in the certificates is

$$e_* = 0.078811099999999995 > 161^{-1/2}.$$

Branch	Boxes	$\widehat{L}$	Spectral counts	Integer range
1	562	< 0	4	$M \geq 161$
2	589	< 0	$N(P) = 6,\ N(P_-) = 4$	$M \geq 161$
3	834	< 0	$N(P) = 8,\ N(P_-) = 4$	$M \geq 161$

Each line is re-proved from its midpoint and radius, including the Krawczyk inclusion, all nondegeneracy conditions, and the coefficient sign. No integer sampling is used for this infinite range. Exact rational endpoint checks prove coverage, and AM2 identifies the whole branch. The first box $[0, 10^{-5}]$ covers every $M \geq 10^{10}$. For $M \leq 160$, the direct reduced-matrix certificate from Section 11 is rerun at every integer: 146, 117 and 74 points, respectively, including its rigorous eigensolver. This proves Theorem D's coefficient and spectral assertions for every relevant integer.

At $\varepsilon = 0$, symbolic evaluation of the same blocks gives

$$\lim_{M\to\infty} \frac{\ell_1(q_1 = 1)}{M} = -\frac{R(R-1)(7R^2 - 34R + 23)}{t(R+1)(5R-3)((R-1)^3 + R + 1)}, \quad R = \sqrt{1+t^2}, \quad t = t_j.$$

To check the identity exactly, substitute $R = (4\tau^4+1)/(4\tau^2)$, $t = (4\tau^4-1)/(4\tau^2)$, $\kappa = (2\tau^2-1)/(2\tau)$, $e^{it} = -(1+it)/R$ in the blocks. Clearing denominators reduces the difference to the zero polynomial (`limit_symbolic.py`). Real-root isolation checks that the normalization and resolvent divisors, including their own denominators, never vanish for $\tau \geq 2$. All the Hopf roots have $\tau > 2.12$. Since $R \geq 65/16 > (17 + 8\sqrt{2})/7$, the displayed coefficient is negative. In particular the first three limits are $-0.00931139838\ldots$, $-0.00794101725\ldots$, $-0.00377769679\ldots$. This is a limit of the discrete-system coefficient, not a deduction from the formal renewal model.

13.7 AM7. Exact onset and the finite-to-infinite junction

The second chart makes the limit $K \to \infty$ regular. Put $s = 1/A$, allow real $M > 3$, and use $\varphi^M = \exp(M \log \varphi)$ in the right half-plane. Define

$$\begin{aligned} f_1 &= \varphi^M + s(\varphi - 1), & f_2 &= s - \varphi + s^{M-1}\varphi^{1-M}, \\ G_1 &= \varphi(s-\varphi)[(1-s)\varphi + (M(1-s)+s)(\varphi-1)], \\ G_2 &= (s-\varphi)[M(\varphi-1) + (1-s)\varphi + \varphi(\varphi-1)], \\ H_1 &= (\varphi-1)[s(1-s) + (M(1-s)+s)(s-\varphi)], \\ H_2 &= (\varphi-1)[(s+M\varphi)(s-\varphi) + s(1-s)\varphi]. \end{aligned}$$

Then $F/A^3 = f_1(H_1 - s^{M-1}\varphi^{-M}H_2) - (G_1 - \varphi^M G_2)f_2$ and $\lambda/A = (1 - s/\varphi)(\varphi - 1)$. These follow by substitution in the boundary determinant defining F. At $s = 0$ the Hopf equations reduce to $u = 0$ and

$$G_\infty(1+iv; M) = 0, \qquad G_\infty(\varphi; M) = -\varphi^{M-2}[\varphi^3 + M(\varphi-1)^2] + (M+1)\varphi - M.$$

The factors s^{M-1} are C^1 at zero. For $0 \leq s \leq b < 1$ and $p = M - 1 > 2$, their values and derivatives are bounded by b^{p_-}, $p_+ b^{p_- - 1}$ and $b^{p_-}|\log b|$, the last provided $b \leq e^{-1/p_-}$.

The script `check_onsets.py` reuses the explicit onset chart from the companion folder, without assuming any result about higher branches. It first encloses (M, u, v) as functions of $s \in [0, 0.0002]$, with $dM/ds > 0$ and $v > 0$. It then encloses (s, u, v) as functions of M up to 162, with $0 < s < 1$ and $v > 0$. Every stored line and every consecutive endpoint is re-proved. A separately enclosed point at the chart transition belongs to both unique-zero boxes; a final sub-box maps inside the last ε-chart box under

$$\varepsilon = M^{-1/2}, \quad \kappa = (s^{-1} - 1)\sqrt{M}, \quad z = M(\varphi - 1).$$

Every integer Hopf point used for $M \leq 161$ is also enclosed afresh and contained in this same curve. The first s-boxes lie in the intervals for M_j^* in Theorem D. Their positive slope, the graph over M, and the nonsingular local Hopf systems show that this arc is the entire connected component: there is no finite interior point where an additional arc can attach. Its two ends leave the domain at $K = \infty$ and $M = \infty$. This proves the exact onsets, including the previously uncertified 87.

Finally, for $3 \leq M \leq 14$, `check_small_stability.py` derives the reduced Jacobian directly from the vector field, computes its characteristic polynomial in $\mathbb{Z}[K][\lambda]$, and verifies positivity on $K > 0$ of every leading Hurwitz determinant. Positive polynomial coefficients or exact real-root isolation establish these inequalities. The Hurwitz criterion proves stability throughout this parameter range.

13.8 AM8. Consequences, verification, and scope

The Hopf theorem now applies at every integer point of the three curves. The sign and crossing direction give the asserted side and stability on the centre manifold. For branch 1 all transverse eigenvalues are stable. For $j = 2, 3$, the monodromy matrix of the small orbit tends to $\exp(2\pi B|_H/\omega)$; a transverse positive-real-part eigenvalue therefore gives a Floquet multiplier outside the unit circle. Positivity of the equilibrium ensures that sufficiently small cycles have positive concentrations.

The complete recheck is `all-M/scripts/recheck_all.py`; its logs, generated onset certificates and direct matrix certificates are in `all-M/verification/recheck-2026-10-25/`. The detailed audit is `all-M/verification.md`. Corrections made during the audit include the fully interval winding count, certified contour containment, and explicit branch junctions. The proof uses correctness of Arb, exact polynomial arithmetic and the standard finite-dimensional Hopf and Krawczyk theorems. It makes no assertion about cycles far from bifurcation, a global stability threshold in κ, or the unresolved higher branches. Those issues are not needed for the complete local answer on branches 1–3.

References

[1] Yuri A. Kuznetsov. Andronov–Hopf bifurcation. *Scholarpedia*, **2006**.
DOI: 10.4249/scholarpedia.1858 2

[2] Robert L. Pego and Juan J. L. Velázquez. Temporal oscillations in Becker–Döring equations with atomization. *Nonlinearity*, **2020**.
DOI: 10.1088/1361-6544/ab6815 arXiv: 1905.02605

[3] Yuri A. Kuznetsov. Elements of Applied Bifurcation Theory. 3rd ed. Springer, **2004**.
DOI: 10.1007/978-1-4757-3978-7

[4] Rudolf Krawczyk. Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken. *Computing*, **1969**.

[5] Ramon E. Moore. A test for existence of solutions to nonlinear systems. *SIAM Journal on Numerical Analysis*, **1977**.

[6] Fredrik Johansson. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, **2017**.
DOI: 10.1109/TC.2017.2690633